Y1, CAT, Marking scheme, Term 1, 2025-26

1. A physical quantity is a property of matter that can be measurable and be assigned a value and unit. 1

   examples: mass (M), length (L) 1

2. b) 2

3. $x = 2.35\,m \pm 0.01\,m$ 2

4. a) $72\,km/h = 72 \times 1000\,m / 3600\,s$

   $= \frac{720}{36}\,m/s$

   $= 20\,m/s$ 2

   b) $25°C = (25 + 273.15)K = 298.15\,K$ 2

   c) $25°C = \left(\frac{9}{5} \times 25 + 32\right)°F = 77°F$ 2

5. a) 2

6. $\cos\theta = \frac{adj}{hyp}$ 0.5 ✓ || $hyp = \overline{RP} = \sqrt{20^2 + 12^2}\,cm$

   $= 23.32$ 0.5 ✓

So $\cos\theta = \frac{12}{23.32} = 0.51$

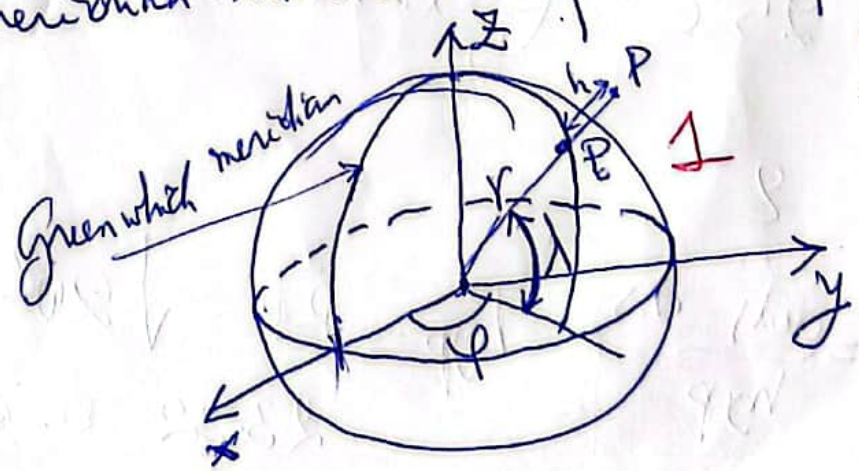
$\theta = \arccos(0.51) = 59.3^\circ$ 1

7. b) 2

8. a) iv) 2

b) a) 2

9. The latitude and longitude are the geographical coordinates where latitude is the angle between the equatorial plane and a line joining the point on the reference ellipsoid to the center of the ellipsoid. The longitude is the angle between x-axis in the equatorial plane and the projection of line joining the origin to where the local meridian intersect the equatorial plane.



1

$\lambda \equiv$ latitude

$\varphi \equiv$ longitude

$h \equiv$ altitude

$-1.9^\circ = ?$

$0.9 \times 60 = 54$

So $-1.9^\circ = 1^\circ 54'$

$0.7 \times 60 = 42$

So $32.7^\circ = 32^\circ 42'$

3

10. b)

3

therefore $(-1.9^\circ, 32.7^\circ) \equiv (-1^\circ 54', 32^\circ 42')$

11. $\vec{u}_\rho = \cos\phi\,\vec{i} + \sin\phi\,\vec{j}$

$\vec{u}_\phi = -\sin\phi\,\vec{i} + \cos\phi\,\vec{j}$

$\vec{u}_z = \vec{k}$

3

12. $G = \dfrac{F r^2}{m_1 m_2}$

$[G] = \dfrac{[F][r^2]}{[m_1][m_2]} = \dfrac{[M][L][T]^{-2}[L]^2}{[M]^2}$

$= [M]^{-1}[L]^3[T]^{-2}$

2

$[G] = \dfrac{m^3}{kg\,s^2}$

b) b) 2

13. $\|A\| = \sqrt{3^2+4^2}\ m = 5\ m$

a) 2

14. $\bar{x} = (19.8+20.3+20.5+20.4+20.6)/5 = 20.3$ 4

absolute Error $E_r = |x-\bar{x}| = |20.2 - 20.3| = 0.1$

percentage error $E_p = \frac{E_r}{x} \times 100\% = \frac{0.1}{20.2} \times 100\%$

$E_p = 0.5\%$

15. i) $f(x) = \sin x$ so $f'(x) = \cos x$

ii) $f(x) = (\sin(2x))^2$ so $f'(x) = 2\cdot(2x)'\cos 2x$

$f'(x) = 4\cos 2x$ 2

16. $r = 5$

$\theta = 60°$ 2

$x = r\cos\theta = 5\cos 60° = \frac{5}{2} = 2.5$

$y = r\sin\theta = 5\sin 60° = 5 \times \frac{\sqrt{3}}{2} = 4.3$

so $(5, 60°) = (2.5, 4.3)$

CamScanner

b) # the python code that converts the polar coordinates onto Cartesian

```
import numpy as np
r = 5
theta = 60
theta_r = np.radians(theta)
x = r * np.cos(theta_r)
y = r * np.sin(theta_r)
print f"r = {r}, theta = {theta} = x = {x}, y = {y}")
```

3

17. a) $\rho = 3$

$\varphi = \pi/4$

$z = 5$

$x = \rho\cos\varphi = 3\cos\frac{\pi}{4} = 3 \times \frac{\sqrt{2}}{2} = 2.1$

$y = \rho\sin\varphi = 3\sin\frac{\pi}{4} = 3 \times \frac{\sqrt{2}}{2} = 2.1$

4

b)

```
import math
# get cylindrical coordinates
rho = float(input("Enter rho:"))
phi = float(input("Enter phi:"))
z = float(input("Enter z"))
# Convert phi onto radians
phir = math.radians(phi)
```

4

18. $\vec{A} = 3\vec{i} + 4\vec{j} - \vec{k}$

$\vec{B} = 2\vec{i} - \vec{k}$

a) $\vec{A} + \vec{B} = 5\vec{i} + 4\vec{j} - 2\vec{k}$ 2

b) $\vec{A} \cdot \vec{B} = 3 \times 2 + 4 \times 0 + (-1)(-1) = 7$ 2

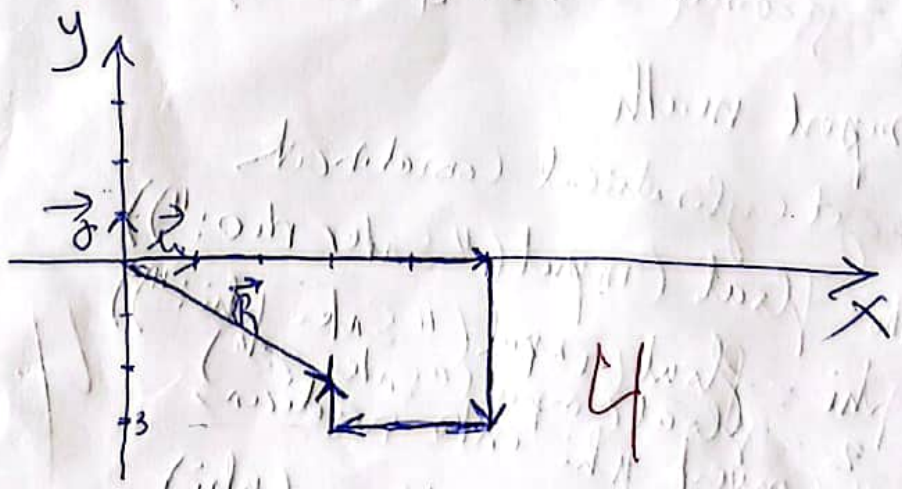
c) $\vec{A} \times \vec{B} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 4 & -1 \\ 2 & 0 & -1 \end{vmatrix} = -4\vec{i} - \vec{j} - 8\vec{k}$ 2

d) $\cos\theta = \dfrac{\vec{A} \cdot \vec{B}}{\|A\| \, \|B\|} = \dfrac{7}{\sqrt{9+16+1} \cdot \sqrt{4+1}} = 0.61$

$\cos\theta = 0.61$ 2

$\theta = 53^\circ$

19.

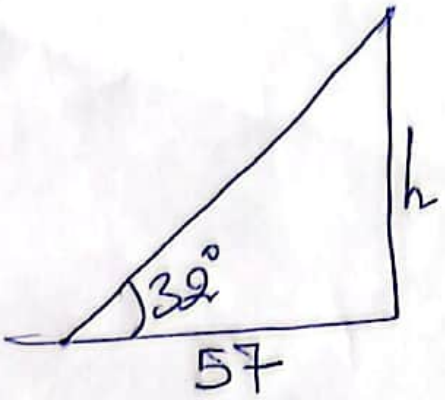


4

$\vec{R} = 3\vec{i} - 2\vec{j}$ 2

the car has displace 3km East and 2 km South 2

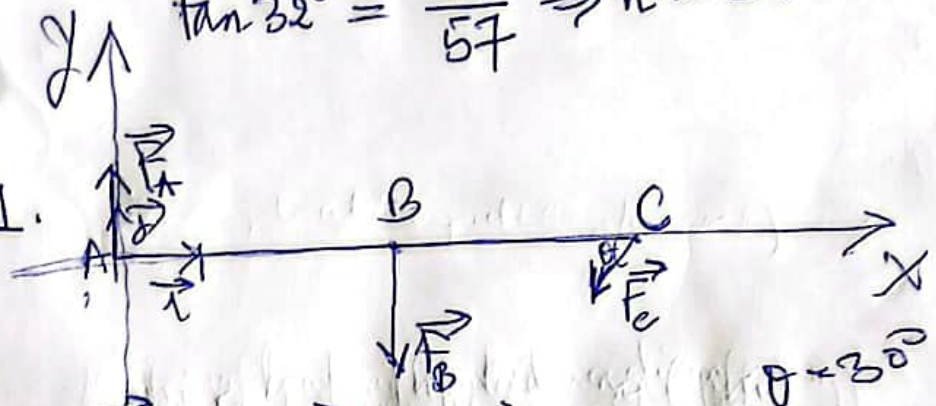
20.



8

$\tan 32° = \frac{h}{57} \Rightarrow h = 57 \times \tan 32° = 35.6\,m$

21.



$\theta = 30°$

$\vec{F_A} = F_A \vec{j} = 10\vec{j}$

$\vec{F_B} = -F_B \vec{j} = -10\vec{j}$

$\vec{F_C} = -F_C \cos 30° \vec{i} - F_C \sin 30° \vec{j}$

$= -20 \times \frac{\sqrt{3}}{2} \vec{i} - 10\vec{j}$

8

$= -10\sqrt{3}\vec{i} - 10\vec{j}$

$= -17.3\vec{i} - 10\vec{j}$

$\vec{F_C} = -17\vec{i} - 10\vec{j}$

$AB = BC = 20\,cm = 0.2\,m$

$\vec{M_C} = \vec{CB} \wedge \vec{F_B} + \vec{CA} \wedge \vec{F_A}$

$= -0.2\vec{i} \wedge (-10\vec{j}) - 0.4\vec{i} \wedge 10\vec{j}$

$= 2\vec{k} - 4\vec{k}$

$\vec{M_C} = -2\vec{k}$